

EE 315 Assignment 5 Report

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1 Introduction

This report presents the implementation and verification of a generalized Gauss–Seidel load-flow program developed for EE 315 Assignment 5.

The program performs load-flow analysis of a power system using the Gauss–Seidel iterative method and determines the bus voltages, bus power, power flow through the transmission lines, and total transmission losses.

The representative four-bus problem (Example 2.7) from the lecture material is used to verify the MATLAB implementation. The program forms the bus admittance matrix Y_{bus} , classifies the buses as slack, PV and PQ buses, performs the Gauss–Seidel iterations with an acceleration factor, and calculates the final power-flow quantities.

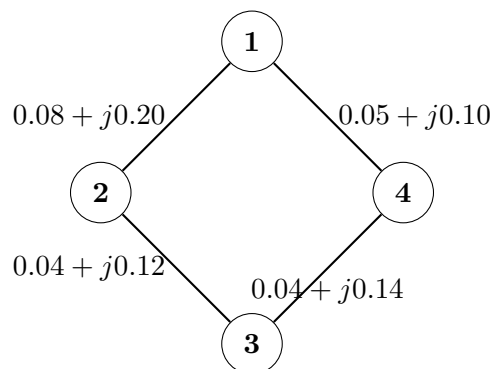
The lecture material defines the slack bus by specified voltage magnitude and angle, the PQ bus by specified real and reactive power, and the PV bus by specified real power and voltage magnitude. The Gauss–Seidel method updates the bus voltages iteratively until the specified convergence criterion is satisfied.

2 Problem: Load Flow of a Four-Bus System

Question

For the sample four-bus system given in the lecture material, determine the bus voltages and bus angles using the Gauss–Seidel iterative method. The system contains one slack bus, one PV bus and two PQ buses.

The system topology consists of the following transmission lines:



Formulae Used

The series admittance of a transmission line is

$$y_{pq} = \frac{1}{R_{pq} + jX_{pq}}. \quad (1)$$

The bus-current equation is

$$\mathbf{I} = \mathbf{Y}_{\text{bus}} \mathbf{V}. \quad (2)$$

For a PQ bus, the Gauss–Seidel voltage update equation is

$$V_i^{(r+1)} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{\left(V_i^{(r)}\right)^*} - \sum_{k=1}^{i-1} Y_{ik} V_k^{(r+1)} - \sum_{k=i+1}^{N_B} Y_{ik} V_k^{(r)} \right]. \quad (3)$$

For a PV bus, the reactive power is first calculated from

$$Q_i = -\text{Im} \left\{ V_i^* \sum_{k=1}^{N_B} Y_{ik} V_k \right\}. \quad (4)$$

The voltage acceleration is performed using

$$V_i^{\text{new}} = V_i^{(r)} + \alpha \left(V_i^{\text{new}} - V_i^{(r)} \right). \quad (5)$$

The iterative process is continued until

$$\max_i \left| V_i^{(r+1)} - V_i^{(r)} \right| < \epsilon. \quad (6)$$

These equations follow the Gauss–Seidel formulation given in the lecture material. [?]

3 Program Inputs

The bus and transmission-line data used in the MATLAB program are given below. These correspond to the four-bus example used for verification.

Table 1: Bus data

Bus	P (pu)	Q (pu)	$ V $ (pu)	Type
1	–	–	1.05	Slack
2	-0.45	–	1.00	PV
3	-0.51	-0.25	–	PQ
4	-0.60	-0.30	–	PQ

Table 2: Transmission-line data

Line	Bus pair	Impedance (pu)
1	1–2	$0.08 + j0.20$
2	1–4	$0.05 + j0.10$
3	2–3	$0.04 + j0.12$
4	3–4	$0.04 + j0.14$

The parameters used for the iterative solution are

$$\alpha = 1.2 \quad (7)$$

and

$$\epsilon = 10^{-5}. \quad (8)$$

The maximum number of iterations is set to 100.

4 Program Result

The MATLAB program converged after 11 Gauss–Seidel iterations.

The convergence history obtained from the program is shown in Table 3.

Table 3: Gauss–Seidel convergence history

Iteration	Maximum Voltage Change
1	6.3254×10^{-2}
2	4.8553×10^{-2}
3	3.1661×10^{-2}
4	9.6379×10^{-3}
5	3.7002×10^{-3}
6	1.3141×10^{-3}
7	4.9338×10^{-4}
8	1.8263×10^{-4}
9	6.8019×10^{-5}
10	2.5322×10^{-5}
11	9.4331×10^{-6}

Since

$$9.4331 \times 10^{-6} < 10^{-5},$$

the convergence criterion is satisfied.

Hence,

$$\boxed{\text{Number of iterations} = 11} \quad (9)$$

and the Gauss–Seidel method is considered converged.

5 Verification

Bus Admittance Matrix

The program calculates the following bus admittance matrix:

$$Y_{\text{bus}} = \begin{bmatrix} 5.724138 - j12.310345 & -1.724138 + j4.310345 & 0 & -4 + j8 \\ -1.724138 + j4.310345 & 4.224138 - j11.810345 & -2.5 + j7.5 & 0 \\ 0 & -2.5 + j7.5 & 4.386792 - j14.103774 & -1.886792 + j6.603774 \\ -4 + j8 & 0 & -1.886792 + j6.603774 & 5.886792 - j14.603774 \end{bmatrix}. \quad (10)$$

This matrix agrees with the Y_{bus} obtained from the transmission-line data.

Converged Bus Voltages

The final bus voltages obtained from the MATLAB program are shown in Table 4.

Table 4: Converged bus voltages

Bus	Type	V_i (pu)	$ V_i $ (pu)	δ_i (deg)
1	Slack	$1.050000 + j0.000000$	1.050000	0.000000
2	PV	$0.992538 - j0.121932$	1.000000	-7.003622
3	PQ	$0.949225 - j0.125784$	0.957522	-7.548417
4	PQ	$0.965385 - j0.076411$	0.968405	-4.525564

The final voltage magnitudes and angles are therefore

$$V_1 = 1.050000 \angle 0^\circ, \quad (11)$$

$$V_2 = 1.000000 \angle (-7.003622^\circ), \quad (12)$$

$$V_3 = 0.957522 \angle (-7.548417^\circ), \quad (13)$$

$$V_4 = 0.968405 \angle (-4.525564^\circ). \quad (14)$$

Bus Power Results

The calculated bus powers are

Table 5: Final bus powers

Bus	Type	$ V $ (pu)	P (pu)	Q (pu)
1	Slack	1.050000	1.653106	0.429161
2	PV	1.000000	-0.449961	0.335138
3	PQ	0.957522	-0.510002	-0.249994
4	PQ	0.968405	-0.600005	-0.299999

Slack-Bus Power

The calculated slack-bus power is

$$P_1 = 1.65310611 \text{ pu} \quad (15)$$

and

$$Q_1 = 0.42916128 \text{ pu}. \quad (16)$$

Thus,

$$S_1 = 1.65310611 + j0.42916128 \text{ pu} \quad (17)$$

Line Flows

The complex power flowing from bus i to bus k is calculated using

$$S_{ik} = V_i [(V_i^* - V_k^*) y_{ik}^*]. \quad (18)$$

The resulting line flows are given in Table 6.

Table 6: Transmission-line power flows

From	To	P_{ij}	Q_{ij}	P_{ji}	Q_{ji}
1	2	0.655873	0.039324	-0.624547	0.038992
1	4	0.997233	0.389837	-0.945240	-0.285851
2	3	0.174586	0.296147	-0.169858	-0.281965
3	4	-0.340143	0.031970	0.345235	-0.014148

Transmission Losses

The real and reactive power losses for each transmission line are calculated by adding the power flows in both directions:

$$S_{\text{loss},ij} = S_{ij} + S_{ji}. \quad (19)$$

The calculated losses are shown below.

Table 7: Transmission-line losses

From	To	P_{loss} (pu)	Q_{loss} (pu)
1	2	0.03132632	0.07831580
1	4	0.05199306	0.10398611
2	3	0.00472732	0.01418195
3	4	0.00509220	0.01782270
Total		0.09313889	0.21430656

Therefore,

$$P_{\text{loss}} = 0.09313889 \text{ pu} \quad (20)$$

and

$$Q_{\text{loss}} = 0.21430656 \text{ pu.} \quad (21)$$

Power-Balance Verification

The total generation obtained from the solution is

$$P_G = 1.65310611 \text{ pu.} \quad (22)$$

The total load is

$$P_L = 1.55996722 \text{ pu.} \quad (23)$$

The calculated loss is

$$P_{\text{loss}} = 0.09313889 \text{ pu.} \quad (24)$$

Hence,

$$P_G - P_L = 1.65310611 - 1.55996722 \quad (25)$$

$$= 0.09313889 \text{ pu.} \quad (26)$$

Therefore,

$$\boxed{P_G - P_L = P_{\text{loss}}} \quad (27)$$

The numerical power-balance error reported by MATLAB is

$$\boxed{5.551 \times 10^{-17} \text{ pu}} \quad (28)$$

which is effectively zero.

6 Overall Result

The main results obtained from the MATLAB program are summarized in Table 8.

Table 8: Overall comparison and verification results

Quantity	Program Result	Expected Result
$ V_2 \angle\delta_2$	$1.0000\angle - 7.004^\circ$	$1.0000\angle - 7.004^\circ$
$ V_3 \angle\delta_3$	$0.9575\angle - 7.548^\circ$	$\approx 0.9575\angle - 7.55^\circ$
$ V_4 \angle\delta_4$	$0.9684\angle - 4.526^\circ$	$\approx 0.9684\angle - 4.53^\circ$
P_1	1.6531 pu	$\approx 1.6531 \text{ pu}$
Q_1	0.4292 pu	$\approx 0.4292 \text{ pu}$
P_{loss}	0.09314 pu	$\approx 0.09314 \text{ pu}$
Q_{loss}	0.21431 pu	$\approx 0.2143 \text{ pu}$

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                                KEY RESULTS
Load-flow status                : CONVERGED
Number of iterations            : 11
Final voltage deviation         : 9.433110e-06 pu
Acceleration factor             : 1.20
Convergence tolerance           : 1.0e-05

Slack bus generation:
    P = 1.653106 pu
    Q = 0.429161 pu

Transmission losses:
    P_loss = 0.093139 pu
    Q_loss = 0.214307 pu

Power balance error             : 5.551e-17 pu

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Figure 1: Results

The MATLAB results agree with the corresponding results of the four-bus example in the lecture material. The uploaded reference report also verifies the same Y_{bus} , converged voltages, slack-bus power and line-flow quantities against the lecture solution.

7 Conclusion

A generalized MATLAB program for Gauss–Seidel load-flow analysis was developed and successfully tested using the four-bus system from the lecture material.

The program forms the bus admittance matrix, handles slack, PV and PQ buses, performs the Gauss–Seidel iterative calculations with an acceleration factor, checks convergence, and calculates the bus voltages, bus powers, transmission-line power flows and transmission losses.

For the selected parameters

$$\alpha = 1.2, \quad \epsilon = 10^{-5},$$

the solution converged in 11 iterations.

The final voltage magnitudes were

$$1.05, \quad 1.00, \quad 0.957522, \quad 0.968405 \text{ pu}$$

for Buses 1–4, respectively.

The total real-power transmission loss was

$$\boxed{0.09313889 \text{ pu}}$$

and the total reactive-power loss was

$$\boxed{0.21430656 \text{ pu}}.$$

The power-balance error was approximately zero, confirming the validity of the obtained load-flow solution.

Therefore, the MATLAB implementation successfully performs the required Gauss–Seidel load-flow analysis and reproduces the expected results of the lecture example.

8 References

1. EE 315 Power Systems, “Gauss–Seidel Method” lecture material, IIT Indore.
2. EE 315 Assignment 5, Example 2.7, Four-Bus Load-Flow Problem.

A MATLAB Program

The MATLAB program used for the load-flow analysis is included below.

[**https://github.com/yashwanth-chitturi/gauss-seidel-load-flow**](https://github.com/yashwanth-chitturi/gauss-seidel-load-flow)